EE3A Rivo Computer Organization HW2

1. addi $t0, $s1, 8
   addi $t1, $s2, -5
   sub $s0, $t0, $t1
2. f=c-d;
   e=f<<4;
3. (a)
   $t0=i*4+P=P[i], P[i]=ACC
   (b)
   $t2=j*4+Q=Q[j], $t0=Q[j+1]
   (c)
   ACC+=Q[j+1]
   (d)
   $t0=Q[j-1]
   (e)
   ACC-=Q[j-1]
4. add $t0, $s0, $s0 #$t0=2i
   sll $t0, $t0, 2
   add $t0, $t0, $s4 #$t0=A[2i] address
   lw $t1, 0($t0) #$t1=A[2i]
   lw $t2, -4($t0) #$t2=A[2i-1]
   sub $t0, $t1, $t2 #$t0=A[2i]-A[2i-1]
   sll $t1, $s0, 2
   add $t1, $t1, $s5 #$t1=B[i] address
   sw $t0, 0($t1) #B[i]= A[2i]-A[2i-1]
5. (a)
   $s0=6fffffff(H)= 01101111111111111111111111111111(B)= 1879048191(D)
   $s1=90000002(H)= 10010000000000000000000000000010(B)= -1879048190(D)
   $t7=$s0-$s1=3758096381(D)=DFFFFFD(H)
   (b)
   $s1=-1879048190<-1 =>$s2=1
   $s0 unsigned=1879048191, $s1 unsigned=2415919106>$s0 =>$s3=1
6. (a)
   $s3=10
   Thisway, $t2=49, 5172B[1]=49, $s0=B[0]=9
   (b)
   $t2=49
   (c)
   B[1]=49